



Application Note

Arbitrary Waveform Generation Toolbox

Description and Operation

Table of Contents

Application Note.....	1
Table of Contents.....	2
Introduction	3
Principle of the Arbitrary Waveform Generation using Vantage	4
Important Notes.....	5
How to use it?.....	6
Manual.....	8
Regular Waveform Tab	8
Arbitrary Waveform Tab	8
Waveform file selection	8
Impulse Response	8
Symbol period	9
Encoder model	9
VS1 level Search	10
VS2 level Search for post processing	10
Functions Panel	10
Frequency Response	10
Generate	11
Tri State Pulse	11
Save TW/Trans and Load TW/Trans	11
Example	11
Chirp wave generation using ArbWave Waveform Tab	11
PulseCode generation of multiple waveforms.....	12
Amplitude adjustment for a custom waveform	13

Introduction

This document describes use of the Verasonics Arbitrary Waveform Generation Toolbox, or GUI-based arbitrary waveform generation tool version v1 (GUI-AWE). This document is contained in the file ArbWaveToolbox_V1_Documentation.pdf. The GUI-AWE enables the Verasonics System to transmit arbitrary waveforms defined by the user into the acoustic medium. The term “arbitrary waveform” is used here to mean an “analog” -type of description of a sampled waveform specified by floating-point values. Conceptually, or in mathematical terms, the arbitrary transmit waveform specification is a vector of real numbers.

The Verasonics Vantage transmitter hardware is restricted to energizing the load on each channel (i.e. each transducer element) with sequences of tri-level pulses, at a pulse rate of 250 MHz. The sequence on each channel may be specified independently. The intensity of each tri-level pulse is defined at three possible levels: two antipodal values defined by the high-voltage transmitter bus, and the ground or zero state. The wide-bandwidth electrical spectrum of the transmit waveform sequence is heavily filtered by the bandpass nature of the transducer physics during conversion to the acoustic domain.

The process of converting an analog waveform specification into a tri-state transmitter sequence is referred to as arbitrary waveform encoding.

By taking advantage of the transducer’s inherent filtering, judicious specification of the tri-state transmit sequence will produce an approximation to the user’s desired analog acoustic transmit waveform. The tool uses proprietary techniques to automatically optimize the transmit sequence in order to give the best match to the user’s desired acoustic waveform. The tool gives the user a convenient graphical interface to manage the constituent parameters required for waveform generation, and to evaluate the quality of the predicted acoustic waveform.

In addition to the desired analog transmit waveform specification, this workflow requires that the user provide a suitable specification for the transducer dynamics parameters, in the form of an impulse response. Please see the document ArbWave_UseModels.pdf for further discussion on specifying the transducer impulse response.

A companion tool, called the Command Line Arbitrary Waveform Encoder (CLawe) tool, provides this same functionality, but with higher accuracy. The command line tool is invoked from the MatLab workspace, and takes a greater execution time than does the GUI-AWE for the same problem specification. For more information on the CLawe, please see its user manual, which is the document ArbWaveEncoder_v2_user_manual.pdf.

White paper

The white paper, Vantage_Arbitrary_Waveform_generation_white_paper_Feb.2016, is also provided to describe Arbitrary Waveform Generation capability and discusses benefits of this technology in detail. <http://verasonics.com/resources-downloads/>

Principle of the Arbitrary Waveform Generation using Vantage

The Verasonics Vantage ultrasound transmitter uses a tri-state pulser that allows specification of arbitrary sequences of three voltage levels $[+V, 0, -V]$ at 4 nS clock intervals (250 MHz clock rate). The voltage V is nominally constant for the duration of the pulse, and is the same for all channels, but the tri-state sequences can be programmed independently for each channel. Furthermore, each acquisition event may emit a unique set of pulse sequences. The voltage may also be changed between events, but the power supply needs some transition time for this to occur.

The Vantage system uses several different methods to specify a transmitted waveform. These include a "Parametric" specification using the pulse frequency, pulse duty cycle (used for amplitude modulation), pulse duration (number of half cycles), and polarity, to generate a basic monochromatic pulse; an apodized frequency modulation pulse specification called "Envelope"; and a "Pulse Code" specification that permits a compact definition of an arbitrary waveform.

The sequences may be of arbitrary length, up to a maximum dependent upon the waveform complexity and transmit memory limitations. An internal storage format that compresses the ternary sequence helps economize transmitter memory usage. The transmitter output transition time between any two states is about 3 clock cycles due to output bandwidth limitations. The output voltage is approximately simulated by a 5-clock running average of the tri-state pulse sequence waveform. Consequently, a perfectly incompressible pulse sequence that allows for the 3-clock state transitions and uses all of the transmitter memory will last more than 60 microseconds at 5 MHz (300+ cycles). The longest duration achievable in a single pulse is 10 Mcycles long, or 2 seconds at 5 MHz.

The record length depends on the complexity of the desired waveform and the output frequency, because we use a compression scheme (based on decomposition of the waveform into nested loops of repeated states) in storing the waveform in transmit memory. The simplest answer is to provide an example: at 5 MHz, a completely incompressible waveform that fills transmit memory will last just over 30 microseconds (equivalent of 150 cycles). However, the transmit bandwidth is limited to about 20 MHz, so that some repetition of states will be required, and the minimum length of a fully complex 5 MHz waveform will likely be closer to 90 microseconds (>450 cycles @ 5 MHz, equivalent). A more accurate estimate requires use of the waveform compressor. Note that the receivers are active during transmit, but are saturated if the transmitter is active.

Important Notes

Before starting, please consider the following important notes.

- This tool is only used to generate pulse code for one or multiple desired waveforms, but no performance is addressed here. In addition, The GUI-based arbitrary waveform generation tool (version v1) described here does not provide an interface to the command-line arbitrary waveform encoder (CLAW) tool version v2. The CLAW operation and usage is described in the file: [ArbWaveEncoder_v2_user_manual.pdf](#).
- In Arbitrary Waveform generation, the cycle limit depends on the available computer memory. User may get an “out of memory” error message or no response from MATLAB during the pulse code generation for a long waveform. If the error occurs, please include the waveform, script, and systemInfo in the email sent to the support.
- The amplitude shown in the figure does not represent the real amplitude transmitted from the probe. This tool only generates the pulse code of a specific “Waveform”.
- Use the “Freq Response” function to confirm that the frequency spectrum of a desired waveform is covered by the transducer bandwidth in order to ensure VDAS acceptable performance.
- Only TW.type and TW.PulseCode will be used by VDAS. The Waveform field in the TW structure does not represent the actual acoustic waveform transmitted from the probe. In addition, in the simulation mode, the waveform will be truncated to 20 cycles.
- Because the showTXPD function only supports parametric TW type, the intensity map can be visualized only if the TW structure is generated with sine wave selected
- There is a limit in the hardware descriptor memory for how long one of this waveform table can be. When the user gets this error, it means that it couldn't create a table that is not allowed due to the memory limit. Please try to make the waveform simpler or shorter.

How to use it?

The **Arb_Main.p** is located in the **Tools/ArbWaveToolbox** folder. Run **Arb_Main.p** and then the GUI, as shown below, will be displayed in few seconds. If an error “Cannot use ArbwaveToolbox because the arbwave option is not enabled” is shown, please contact Verasonics to purchase the arbwave option. In the figure below, the numbers 1-4 below indicate the steps of pulse code generation. The Arbitrary Waveform Generation Tool is used to generate the pulse code that is sent to the VDAS system. After generating the pulse code of a desired waveform, the original TW structure needs to be replaced in the SetUp scripts. As shown in the figure, the pulse code can be generated in 4 steps:

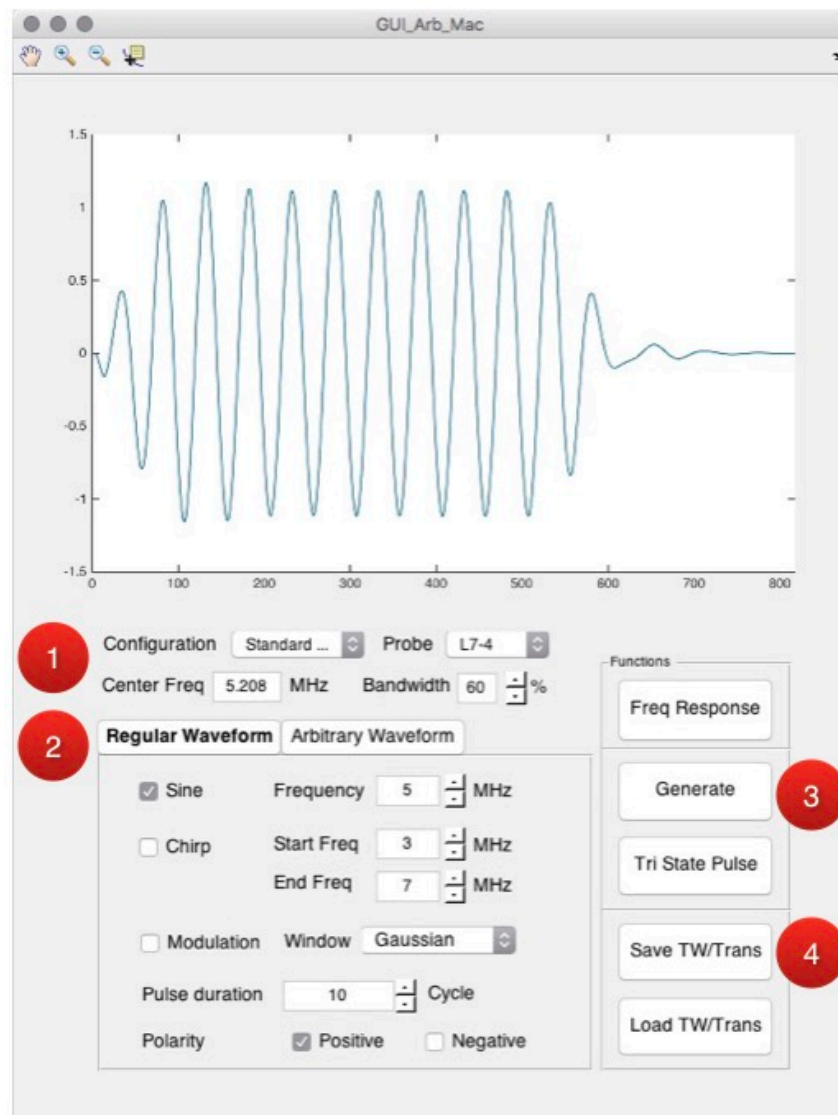


Fig 1. The GUI of the ArbWaveToolbox

1. Configuration, Probe and Bandwidth selection. Select the hardware configuration and probe type. The default bandwidth is 60%. The center frequency is fixed for known probes. Because the impulse response of many probes has not been measured, the bandwidth is flexible so the user can determine the best number for experiments. If the user has a custom probe, please select “Custom” and enter the corresponding center frequency and bandwidth.
2. Regular Waveform or Arbitrary Waveform. In the Regular Waveform tab, the pulse code of a sine wave or a chirp wave with amplitude modulation (AM) will be generated using either a parametric or envelope type of TW structure. To use the Arbitrary Waveform tab, a customized waveform is required, and the corresponding pulse code will be generated using the *least squares signal approximation algorithm* (The definition of each parameter will be introduced below).
3. After determination of all required parameters, click “Generate” for pulse code generation.
4. Save TW/Trans structures to the desired location for VDAS setup.

Next, modify the TW structure definition in the SetUp scripts. For instance, the TW has been generated and saved as “desiredWaveform.mat”. Load this mat file, and the VSX will acquire images using the desired TW structure.

```
% Specify Transmit waveform structure.  
% TW(1).type = 'parametric';  
% TW(1).Parameters = [Trans.frequency,.67,1,1];    % A, B, C, D  
  
load('desiredWaveform','TW');
```

If profile 5, i.e., extended burst option, is required for the waveform, an example script, SetUpL7_4ArbPush, is provided in the ArbWaveToolbox/Examples to demonstrate how to correctly implement profile 5, along with the waveform file, into the SetUp script.

In addition, a tutorial video is also provided on our website, <http://verasonics.com/tools-demonstration/>. Please follow the video for better understanding of the tool.

Manual

Regular Waveform Tab

A sine wave or a chirp wave, with or without AM, can be generated in the regular waveform tab. The “Frequency” indicates the transmit frequency, not the center frequency of transducer. If the transmit frequency is far away from the center frequency, poor efficiency will be expected.

In addition, the Start and End frequency of a chirp wave, three AM types, the number of cycles, and the polarity can be easily applied and the corresponding waveform “FOR VERASONICS SIMULATION” will be displayed in the GUI. Note: In the simulation, the waveform will be truncated to ~20 cycles.

Arbitrary Waveform Tab

One of the three actions will be applied while the “Arbitrary Waveform” tab is selected: 1) If a waveform file has not been selected, a window will pop out for waveform selection. 2) If a waveform file has been selected but the pulse code has not been generated, the waveform will be plotted. 3) If the pulse code of a selected waveform has been generated, the acoustic waveform and corresponding error will be plotted.

The definition of each parameter shown in the tab is defined below and the algorithm is described in Arbitrary waveforms using a tri-state transmit pulser, Flynn, J.A. et. al, Ultrasonics Symposium (IUS), 41-41, 2013. <http://dx.doi.org/10.1109/ULTSYM.2013.0011>

Waveform file selection

Two methods can be used to select waveform file. First, “Choose waveform file...” is used to select a waveform file and the corresponding file path will be shown. Second, copy the file path and paste to the GUI (“waveform.txt” shown in above figure). Here, two formats are supported. One is .mat file in which the waveform vector is the first variable. The toolbox will retrieve the vector in the first variable into the workspace. The other one is .txt file which stores the waveform vector. The waveform will be plotted after successfully retrieving data. Please note that 1) the sampling rate of the waveform should be at 250 MHz. 2) The voltage scale will not affect the pulse code generation.

Impulse Response

Generally, a nominal transducer impulse response computation with butterworth IIR is provided. User can also apply their own impulse response with “Customized” option. The “factory setting” option is only available for L7-4 probe and the impulse response is included in “IRL7_4.mat”. The impulse response should be in time domain, with 250 MHz sampling rate and scale to 100 counts peak-to-peak. After measurement and scale adjustment, please save the impulse response vector as .mat file. In the toolbox, after the selection of “Customized” option, user will be requested to select the correct impulse response file.

Symbol period

The symbol period is typically based on the nominal transducer center frequency. Each symbol is comprised of several transmit clock periods and therefore several tri-state voltage instances. A typical choice of symbol period corresponds to 1/4 the nominal center frequency. For example, the system clock in Vantage is 250 MHz, so the symbol period of the L7-4 probe will be $250/5.208/4 \approx 12$. Additionally, the symbol period selection determines the number of pulse width modulation (PWM) levels available. The hypothetical twelve clock symbol period permits 25 PWM levels, including the zero-voltage level. Figure 2 shows 12 positive PWM levels and corresponding acoustic waveform. The other 12 negative levels can be generated by flipping the sign. One of the 25 levels is used as a prototype to determine the best scale level of the VS1 search. The original amplitude, however, will not be taken into account due to the average calculation for the scale determination.

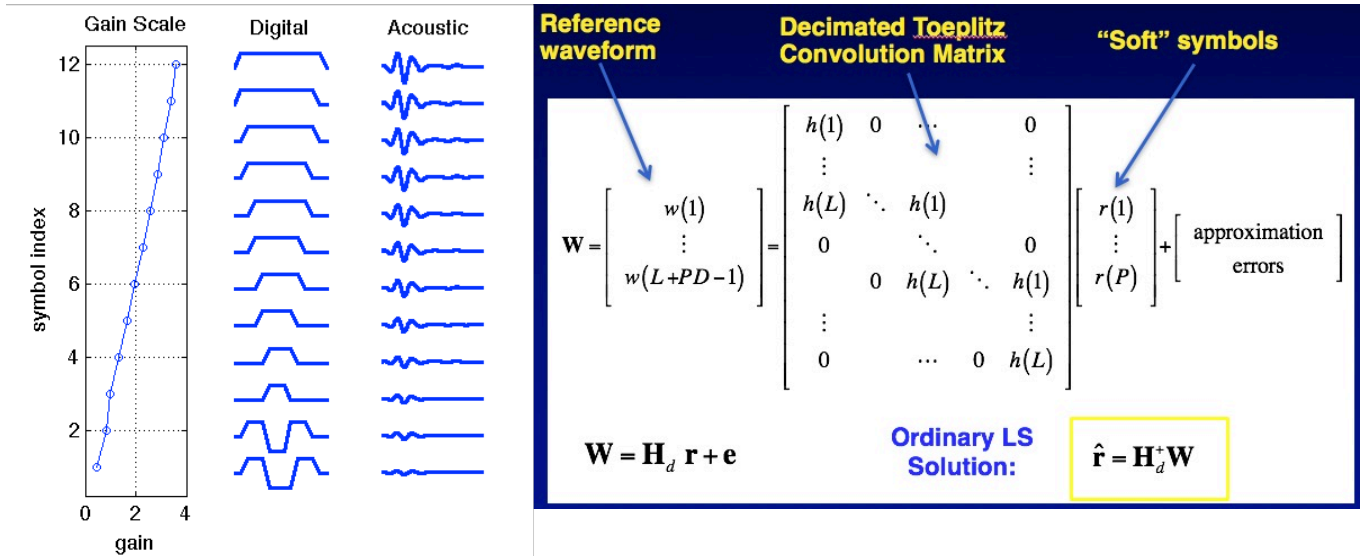


Fig. 2. The illustration of symbol period that is used to determine the number of pulse width modulation (PWM) levels available and the prototype used in the Toeplitz Convolution Matrix. The scale range of the soft symbols will be determined by "VS1 level Search".

Encoder model

The fidelity metric employed by the encoder is a design choice that depends on the usage model of the application. Normalized RMS error metrics considered here measure (1) closeness of the reference waveform's predicted acoustic pressure, and (2) closeness of the reference waveform to acoustic pressure to that achieved by DAC. We denote these the one-way probe compensation and one-way DAC synthesis problems, respectively.

VS1 level Search

In order to determine the best overall quantization level regardless of the voltage of the original waveform, the scale levels should be defined, including the start value, the end value, and the intervals. The toolbox will determine the best level within the defined range. For instance, with the default setting, 200 levels will be generated in a range of 0.2 and 200 using linspace function built in the Matlab. The Soft symbols shown in Fig. 2 will be scaled by the levels and best level will be listed in the final statistical values printed out on the command window.

VS2 level Search for post processing

User may be able to refine the estimation obtained by VS1 search. VS2 search generates new pulse code based on the best level obtained from the VS1 Search. The corresponding soft symbols are scaled by the VS2 Search range and only 3 symbol periods are used in the optimization process. The computation time, thereby, will be longer than VS1 search, especially with more iterations and level intervals. **Note:** the performance of VS2 might not be better than VS1. User can try different parameters to get the best pulse code for a designed waveform.

Functions Panel

As shown in Fig. 1, some functions are listed on the right panel of the GUI. Below is the instruction of these functions.

Frequency Response

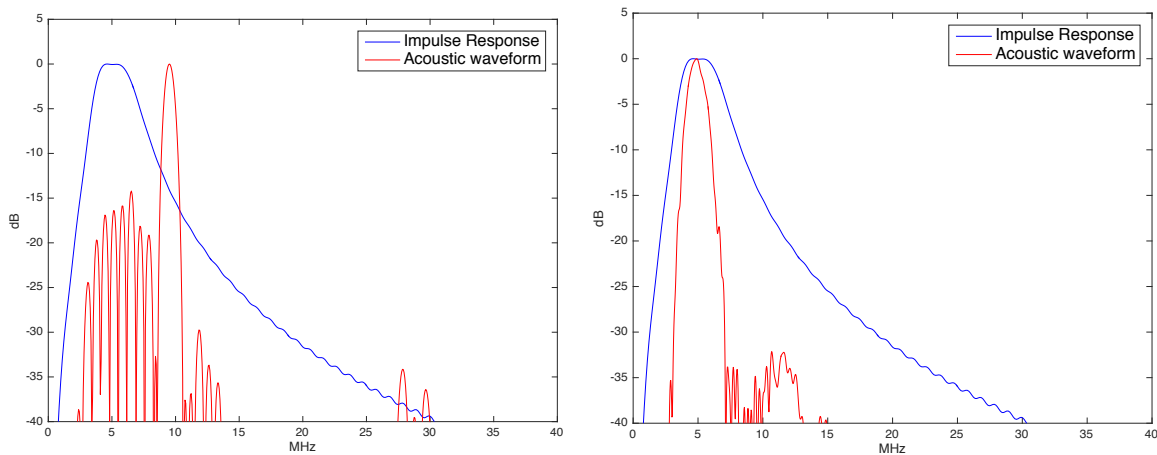


Fig. 3 The frequency response of a 10-MHz sinusoid waveform (left) and a Chirp waveform (3-7 MHz, right).

The frequency spectrum of impulse response and acoustic waveform (in Regular Waveform Tab) or original waveform (in Arbitrary Waveform Tab) can be used for efficiency estimation. As

shown in Fig. 3, the accuracy of the pulse code estimation would be lower if a 10-MHz waveform is transmitted using L7-4 probe.

Generate

Generate is used to generate PulseCode which will be applied to VDAS. The PulseCode will not be saved until the “Save TW/Trans” is clicked. Please pay attention to the figure title which will remind the user to save the pulseCode, as shown in the tutorial video.

Tri State Pulse

Tri State Pulse is stored in TW.triState and is used to visualize the Tri-State Transmit Pulser which will be transmitted to the Probe. For instance, the 0.67 pulse width will be displayed with a sine wave.

Save TW/Trans and Load TW/Trans

Three variables will be saved to the customized location after clicking “Save TW/Trans” , including 1) Trans – transducer specifications, 2) TW – waveform and pulse code, and 3) Handle – the GUI handle information for user to retrieve parameters after clicking “Load TW/Trans” button.

Example

The tutorial video demonstrates how to use the ArbWaveToolbox to generate the pulse code of a sinusoid waveform, a chirp waveform, and a custom waveform. More information is listed below.

Chirp wave generation using ArbWave Waveform Tab

Three illustrative waveform files, Chirp_AM_0.1to1.1MHz.txt, Chirp_AM_3to7MHz.txt, and Chirp_AM_3to7MHz.mat, created by **MakingChirp.m** in the Examples folder, are included in Examples folder. Click Arbitrary Waveform tab and select any waveform file. Since the impulse response of the L7-4 probe is provided, using Factory setting would be recommended. Click Generate for pulse code generation, the statistical values will be printed in the command window and the error value will be shown as the figure below. The red line indicates the error between the estimated and the original waveform. This approach uses knowledge of the transducer impulse response to develop an optimal ternary pulse sequence that reproduces the ideal analog waveform with a residual error of typically between -20 and -30 dB, depending on the transducer characteristics.

Figure 4 shows the final statistical values after VS1 and VS2 search using L7-4 probe and an impulse Response based on a Butterworth bandpass filter. The levels for VS1 search are 0.2 to 2 with 200 levels, as well as the 0.9 to 1 with 5 levels for VS2 search. The delaySpread is the length of the impulse response above the threshold (-30dB). The Error Value of VS2 search is higher than VS1, indicating that the post-process using 3 symbol periods won't always provide better estimation.

```
Final statistical values of VS1 search:
delaySpread | Best level | Error Value
0.96 us      | 1.01         | -24.11 dB

Final statistical values of VS2 search:
              | Best level | Error Value
              | 0.97       | -24.05 dB
```

Fig. 4, Final statistical values

PulseCode generation of multiple waveforms

The **MultiTWPulseCode.m** in the Examples folder is used to generate the pulse code of multiple waveforms. If the user would like to transmit different waveforms for each element, the number of column of the waveform matrix, should be the same as the number of transducer elements. First, have the custom waveform file ready. The correct format could be seen from another waveform file, **TestMultiWaveform.mat**, provided in the same folder. It includes four waveforms generated by the Regular Waveform Tab and the matrix size is [1000 x 4]. (Note: sample rate is 250 MHz)

	Wave	Modulation	Cycles	Polarity
Wave 1	Chirp 3-7 MHz	Gaussian	10	+
Wave 2	Chirp 3-7 MHz	Hanning	10	-
Wave 3	Sinusoid 5 MHz	-	5	+
Wave 4	Sinusoid 5 MHz	Hanning	10	-

Second, define required parameters:

- Transducer: As regular SetUp script, Trans. name and Trans.units are required.
- bandwidth: Only used in the butterworth IR generation
- IR_Option: Three options, as shown in the Arb_GUI, are provided. 1 is nominal butterworth, 2 is customized, and 3 is Factory Setting. The factory impulse response is provided with L7-4 probe only (IRL7_4.mat)
- The Encoder Model, VS1 and VS2 search.

Third, run the script and select correct waveform file. The final statistical values will also be displayed.

Amplitude adjustment for a custom waveform

The ArbWaveToolbox will perform estimation and normalization based on the "level search range" defined in the GUI. For instance, the default VS1 level is 0.2 to 2 with 200 levels, which is the scale level of the "Soft" Symbols shown in Fig. 2. If the desired waveform with half amplitude is the goal, the user should obtain the best level for the desired waveform and set the level range around the half of it. For instance, if the best level of the full amplitude waveform is 0.8, using 0.35 to 0.45 with 100 levels might be able to get the PulseCode for half amplitude of the waveform.

However, the performance may decrease and the nonlinearity may be higher. The adjustment of Receive.Apod might be required for parameters optimization. More levels might have better estimation for half amplitude. Again, having impulse response would lead to better estimation.

*This document is intended for use by Verasonics customers only.
Do not duplicate or distribute without permission.*

Verasonics, Inc.

11335 NE 122nd Way, Suite 100,

Kirkland, WA 98034

USA